

ApexPlanet — Task 4

Exploitation & System Security

Controlled Metasploitable2 Lab Assessment

Assessment target: Metasploitable2 — 192.168.56.102

Assessment VM: Kali Linux — 192.168.56.101 (lab interface)

Scope: Private VirtualBox lab environment

Executive Summary

This report documents a controlled assessment of the intentionally vulnerable Metasploitable2 virtual machine. The evidence covers reconnaissance, service/version discovery, vulnerability assessment, SSH security review, system-hardening checks, phishing awareness simulation, and a benign malware-analysis exercise.

The clearest documented vulnerability is VSFTPD 2.3.4. Metasploit was used to identify the corresponding VSFTPD 2.3.4 backdoor module and review its configuration. The captured evidence does not claim successful remote-shell or post-exploitation access.

1. Scope and Methodology

The assessment was performed only against the deliberately vulnerable lab target on the private VirtualBox network. The workflow consisted of host discovery, service/version enumeration, vulnerability identification, configuration review, hardening-state checks, and controlled awareness/analysis exercises.

Lab addresses observed in the evidence

System	Address	Role
Kali Linux	192.168.56.101	Assessment VM
Metasploitable2	192.168.56.102	Intentionally vulnerable target

2. Reconnaissance and Service Discovery

Host discovery located the Metasploitable2 VM on the lab subnet. A version/OS scan then identified a broad exposed service surface, including FTP, SSH, Telnet, SMTP, DNS, HTTP, SMB, RPC, databases, VNC, IRC, AJP and Tomcat.

Reconnaissance / host discovery

```
kali@kali: ~  
Session Actions Edit View Help  
Docs: man:ssh(8)  
man:ssh_config(5)  
--(kali@kali)-~  
_ _ _  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-11 05:09 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.013s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
23/tcp    open  telnet       Linux telnetd  
25/tcp    open  smtp         Postfix smtpd  
53/tcp    open  domain       ISC BIND 9.4.2  
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
111/tcp   open  rpcbind      2 (RPC #100000)  
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
512/tcp   open  exec         netkit-rsh rshexec  
513/tcp   open  login        Netkit rshd  
514/tcp   open  shell        Netkit rshd  
1099/tcp  open  java-rmi     GNU Classpath gmrregistry  
1524/tcp  open  bindshell    Metasploit3 root shell  
2049/tcp  open  nfs          2-4 (RPC #100003)  
2121/tcp  open  ftp          ProFTPD 1.3.1  
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5  
5432/tcp  open  postgresql   PostgreSQL 8.3.0 - 8.3.7  
5900/tcp  open  vnc           VNC (protocol 3.3)  
6000/tcp  open  X11          (access denied)  
6667/tcp  open  irc          UnrealIRCd  
8080/tcp  open  ajp13        Apache jkerv (Protocol v1.3)  
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1  
MAC Address: 08:00:27:82:32:8E (Oracle VirtualBox virtual NIC)  
Device type: general purpose  
Running: Linux 2.6.x  
OS CPE: cpe:/o:linux:linux_kernel:2.6  
OS details: Linux 2.6.9 - 2.6.33  
Network Distance: 1 hop  
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OS: Unix, Linux; CPE: cpe:/o:linux:linux_kernel  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 15.59 seconds  
--(kali@kali)-~  
_ _ _
```

Service and version discovery overview

```
kali@kali: ~  
Session Actions Edit View Help  
| DF059135-2CF5-5441-8F22-E6EF1DEE5F6E 10.0 https://vulners.com/gitee/DF059135-2CF5-5441-8F22-E6EF1DEE5F6E *EXPLOIT*  
| 26088200-7E01-5A22-AD21-20F63E8EC7CE 10.0 https://vulners.com/githubexploit/26088200-7E01-5A22-AD21-20F63E8EC7CE *EXPLOIT*  
| PACKETSTORM:173661 9.8 https://vulners.com/packetstorm/PACKETSTORM:173661 *EXPLOIT*  
| F0979183-AE88-538A-86CF-3AF0523F3807 9.8 https://vulners.com/githubexploit/F0979183-AE88-538A-86CF-3AF0523F3807 *EXPLOIT*  
| CVE-2023-38408 9.8 https://vulners.com/cve/CVE-2023-38408 *EXPLOIT*  
| CVE-2016-1908 9.8 https://vulners.com/cve/CVE-2016-1908 *EXPLOIT*  
| CVE-2018-4478 9.8 https://vulners.com/cve/CVE-2018-4478 *EXPLOIT*  
| CEF8C43-8FE2-585E-8918-54A900A09284 9.8 https://vulners.com/githubexploit/CEF8C43-8FE2-585E-8918-54A900A09284 *EXPLOIT*  
| B8190CDB-3EB9-5631-9828-8064A1575823 9.8 https://vulners.com/githubexploit/B8190CDB-3EB9-5631-9828-8064A1575823 *EXPLOIT*  
| A2836885-C737-548F-8C0A-9339EDCDBFF5 9.8 https://vulners.com/githubexploit/A2836885-C737-548F-8C0A-9339EDCDBFF5 *EXPLOIT*  
| 8FC9C8A8-3980-5F3C-8D5E-E8085379A623 9.8 https://vulners.com/githubexploit/8FC9C8A8-3980-5F3C-8D5E-E8085379A623 *EXPLOIT*  
| 8A001159-548E-544E-AA87-20E89F397EC 9.8 https://vulners.com/githubexploit/8A001159-548E-544E-AA87-20E89F397EC *EXPLOIT*  
| 6192C35D-F788-5C0A-AB8D-9826A79A5320 9.8 https://vulners.com/githubexploit/6192C35D-F788-5C0A-AB8D-9826A79A5320 *EXPLOIT*  
| 2227729D-6700-5C8F-8938-1EEAFD4B9FF0 9.8 https://vulners.com/githubexploit/2227729D-6700-5C8F-8938-1EEAFD4B9FF0 *EXPLOIT*  
| 0221525F-07F5-5790-912D-F489E2018507 9.8 https://vulners.com/githubexploit/0221525F-07F5-5790-912D-F489E2018507 *EXPLOIT*  
| CVE-2026-60002 9.4 https://vulners.com/cve/CVE-2026-60002 *EXPLOIT*  
| CVE-2015-5600 8.5 https://vulners.com/cve/CVE-2015-5600 *EXPLOIT*  
| CVE-2026-35414 8.1 https://vulners.com/cve/CVE-2026-35414 *EXPLOIT*  
| CVE-2026-35385 8.1 https://vulners.com/cve/CVE-2026-35385 *EXPLOIT*  
| CVE-2026-35385 8.1 https://vulners.com/cve/CVE-2026-35385 *EXPLOIT*  
| BA3887BD-F579-53B1-AA4A-FF49E9531C0 8.1 https://vulners.com/githubexploit/BA3887BD-F579-53B1-AA4A-FF49E9531C0 *EXPLOIT*  
| 4F801808-F993-5CAF-BD57-D7E290D10C1F 8.1 https://vulners.com/githubexploit/4F801808-F993-5CAF-BD57-D7E290D10C1F *EXPLOIT*  
| PACKETSTORM:140070 7.8 https://vulners.com/packetstorm/PACKETSTORM:140070 *EXPLOIT*  
| PACKETSTORM:101052 7.8 https://vulners.com/packetstorm/PACKETSTORM:101052 *EXPLOIT*  
| EXPLOITPACK:5BCA798C6BA7FAE29334297EC0B6A09 7.8 https://vulners.com/exploitpack/EXPLOITPACK:5BCA798C6BA7FAE29334297EC0B6A09 *EXPLOIT*  
| E08-ID:40888 7.8 https://vulners.com/exploitdb/E08-ID:40888 *EXPLOIT*  
| CVE-2020-15778 7.8 https://vulners.com/cve/CVE-2020-15778 *EXPLOIT*  
| CVE-2016-6515 7.8 https://vulners.com/cve/CVE-2016-6515 *EXPLOIT*  
| CVE-2016-10012 7.8 https://vulners.com/cve/CVE-2016-10012 *EXPLOIT*  
| CVE-2015-8325 7.8 https://vulners.com/cve/CVE-2015-8325 *EXPLOIT*  
| C9A132FD-1FA5-53A2-86EE-0DAF45EEFF3 7.8 https://vulners.com/githubexploit/C9A132FD-1FA5-53A2-86EE-0DAF45EEFF3 *EXPLOIT*  
| C892A90E-C1D1-SAS4-BFAA-046266448553 7.8 https://vulners.com/githubexploit/C892A90E-C1D1-SAS4-BFAA-046266448553 *EXPLOIT*  
| 99C4CA40-30C8-SA3A-83A3-4B5E7A9E168A 7.8 https://vulners.com/githubexploit/99C4CA40-30C8-SA3A-83A3-4B5E7A9E168A *EXPLOIT*  
| 991D2CC4-0E09-5745-97A2-4917461B0BEC 7.8 https://vulners.com/githubexploit/991D2CC4-0E09-5745-97A2-4917461B0BEC *EXPLOIT*  
| 4F3BA0CA-CAD8-53FA-A857-3EAC9C4C3443 7.8 https://vulners.com/githubexploit/4F3BA0CA-CAD8-53FA-A857-3EAC9C4C3443 *EXPLOIT*  
| 3121653-7FD9-5769-BD0A-4129BE911406 7.8 https://vulners.com/githubexploit/3121653-7FD9-5769-BD0A-4129BE911406 *EXPLOIT*  
| 2E719186-2FED-58A8-A150-762EFBAA523 7.8 https://vulners.com/gitee/2E719186-2FED-58A8-A150-762EFBAA523 *EXPLOIT*  
| 23C097BE-7C95-513B-9E73-298C48D74432 7.8 https://vulners.com/githubexploit/23C097BE-7C95-513B-9E73-298C48D74432 *EXPLOIT*  
| 13370AY-ID-26494 7.5 https://vulners.com/cve/13370AY-ID-26494 *EXPLOIT*  
| 10213DBE-F683-S8BB-8603-353173626207 7.8 https://vulners.com/githubexploit/10213DBE-F683-S8BB-8603-353173626207 *EXPLOIT*  
| SSV:92579 7.5 https://vulners.com/seebug/SSV:92579 *EXPLOIT*  
| SSV:61450 7.5 https://vulners.com/seebug/SSV:61450 *EXPLOIT*  
| E08-ID:40963 7.5 https://vulners.com/exploitdb/E08-ID:40963 *EXPLOIT*  
| CVE-2026-60000 7.5 https://vulners.com/cve/CVE-2026-60000 *EXPLOIT*  
| CVE-2026-59999 7.5 https://vulners.com/cve/CVE-2026-59999 *EXPLOIT*  
| CVE-2016-10708 7.5 https://vulners.com/cve/CVE-2016-10708 *EXPLOIT*  
| CVE-2015-10009 7.5 https://vulners.com/cve/CVE-2015-10009 *EXPLOIT*  
| CVE-2014-1692 7.5 https://vulners.com/cve/CVE-2014-1692 *EXPLOIT*  
| CVE-2010-5107 7.5 https://vulners.com/cve/CVE-2010-5107 *EXPLOIT*  
| CFS2FA19-B5D8-5014-B50F-2411851976E2 7.5 https://vulners.com/githubexploit/CFS2FA19-B5D8-5014-B50F-2411851976E2 *EXPLOIT*  
| 13370AY-ID-26576 7.5 https://vulners.com/cve/13370AY-ID-26576 *EXPLOIT*  
| SSV:92582 7.2 https://vulners.com/seebug/SSV:92582 *EXPLOIT*  
| E08-ID:40962 7.0 https://vulners.com/exploitdb/E08-ID:40962 *EXPLOIT*
```

Vulnerability assessment evidence

```
kali@kali: ~  
Session Actions Edit View Help  
Metasploit Documentation: https://docs.metasploit.com/  
The Metasploit Framework is a Rapid7 Open Source Project  
msf > nmap -sV --script vuln 192.168.56.102  
[*] exec: nmap -sV --script vuln 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-11 05:20 -0400  
Nmap Scan Report for 192.168.56.102  
Host is up (0.038s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp           vsftpd 2.3.4  
| vulners:  
| vsftpd 2.3.4:  
| PACKETSTORM-102145 10.0 https://vulners.com/packetstorm/PACKETSTORM-102145 *EXPLOIT*  
| MSF-EXPLOIT-UNIX-FTP-VSFTPD_234_BACKDOOR- 10.0 https://vulners.com/metasploit/MSF-EXPLOIT-UNIX-FTP-VSFTPD_234_BACKDOOR- *EXPLOIT*  
| FE4240DE-72FF-5445-B198-9E205322DC86 10.0 https://vulners.com/githubexploit/FE4240DE-72FF-5445-B198-9E205322DC86 *EXPLOIT*  
| E08-1D149757 10.0 https://vulners.com/exploitdb/E08-1D149757 *EXPLOIT*  
| E08-1D149757 10.0 https://vulners.com/exploitdb/E08-1D149757 *EXPLOIT*  
| E08791E3-8935-5632-A403-CB7E7F2A26C8 10.0 https://vulners.com/githubexploit/E08791E3-8935-5632-A403-CB7E7F2A26C8 *EXPLOIT*  
| EC73033B-7726-59C3-828B-0828B8955089 10.0 https://vulners.com/githubexploit/EC73033B-7726-59C3-828B-0828B8955089 *EXPLOIT*  
| E980AEBB-5138-50BF-8922-2087E3C046D0 10.0 https://vulners.com/githubexploit/E980AEBB-5138-50BF-8922-2087E3C046D0 *EXPLOIT*  
| E356CEC-7C81-5A77-8C4A-87DFA8D21897 10.0 https://vulners.com/githubexploit/E356CEC-7C81-5A77-8C4A-87DFA8D21897 *EXPLOIT*  
| DFC7E45F-6E41-513A-8114-88C1BF0226E1 10.0 https://vulners.com/githubexploit/DFC7E45F-6E41-513A-8114-88C1BF0226E1 *EXPLOIT*  
| D8CFC54B-5E80-55FA-85C1-920406A8FF9F 10.0 https://vulners.com/githubexploit/D8CFC54B-5E80-55FA-85C1-920406A8FF9F *EXPLOIT*  
| D0B0A85F-750F-5F9B-9C69-65E8FA6977D5 10.0 https://vulners.com/githubexploit/D0B0A85F-750F-5F9B-9C69-65E8FA6977D5 *EXPLOIT*  
| CVE-2011-2523 10.0 https://vulners.com/cve/CVE-2011-2523  
| CVD-2028-46837 10.0 https://vulners.com/cvd/CVD-2028-46837  
| CC3F6C15-182F-53F6-ASCC-812D37F1F047 10.0 https://vulners.com/githubexploit/CC3F6C15-182F-53F6-ASCC-812D37F1F047 *EXPLOIT*  
| BEE678EA-45C2-50C8-88F3-9A2E7CC2DECE 10.0 https://vulners.com/githubexploit/BEE678EA-45C2-50C8-88F3-9A2E7CC2DECE *EXPLOIT*  
| A7A98EE8-8AB4-5A6C-B232-E31EB21F6397 10.0 https://vulners.com/githubexploit/A7A98EE8-8AB4-5A6C-B232-E31EB21F6397 *EXPLOIT*  
| A4180BAB-C86F-56F2-89E8-709583C4D553 10.0 https://vulners.com/githubexploit/A4180BAB-C86F-56F2-89E8-709583C4D553 *EXPLOIT*  
| A4185EAD-1A4C-56A6-97C6-1C58A1CF1E3B 10.0 https://vulners.com/githubexploit/A4185EAD-1A4C-56A6-97C6-1C58A1CF1E3B *EXPLOIT*  
| 98A8A3A1-1CBF-59D6-AA48-33596997F238 10.0 https://vulners.com/githubexploit/98A8A3A1-1CBF-59D6-AA48-33596997F238 *EXPLOIT*  
| 9180FCF-4B0D-502E-8527-5E5AC2924A49 10.0 https://vulners.com/githubexploit/9180FCF-4B0D-502E-8527-5E5AC2924A49 *EXPLOIT*  
| 8850985A-A37F-5A73-9F9A-E33F205D5380 10.0 https://vulners.com/githubexploit/8850985A-A37F-5A73-9F9A-E33F205D5380 *EXPLOIT*  
| 88954789-E1E3-5DA3-86E4-387A3D4ACCA8 10.0 https://vulners.com/githubexploit/88954789-E1E3-5DA3-86E4-387A3D4ACCA8 *EXPLOIT*  
| 817C08FE-87CA-5FE8-B982-8CC64333A3F3 10.0 https://vulners.com/githubexploit/817C08FE-87CA-5FE8-B982-8CC64333A3F3 *EXPLOIT*  
| 7AA4C193-6858-5FAC-88A5-EC3B70218822 10.0 https://vulners.com/githubexploit/7AA4C193-6858-5FAC-88A5-EC3B70218822 *EXPLOIT*  
| 71E1A4CA-8A98-556B-AB36-EBC2856780EF 10.0 https://vulners.com/githubexploit/71E1A4CA-8A98-556B-AB36-EBC2856780EF *EXPLOIT*  
| 640E6781-DC00-5CC2-88A5-C12AFFCA478B 10.0 https://vulners.com/githubexploit/640E6781-DC00-5CC2-88A5-C12AFFCA478B *EXPLOIT*  
| 63A5C9A7-C241-5E83-9EE6-ABA8448D0270 10.0 https://vulners.com/githubexploit/63A5C9A7-C241-5E83-9EE6-ABA8448D0270 *EXPLOIT*  
| 5F8ACEDC-770F-5054-851A-0A38B76A5009 10.0 https://vulners.com/githubexploit/5F8ACEDC-770F-5054-851A-0A38B76A5009 *EXPLOIT*  
| 5F8ACAF8-8FE6-54B0-9162-3EE88D090600 10.0 https://vulners.com/githubexploit/5F8ACAF8-8FE6-54B0-9162-3EE88D090600 *EXPLOIT*  
| 598AFDCD-5027-5C82-BC59-FC76250081D8 10.0 https://vulners.com/githubexploit/598AFDCD-5027-5C82-BC59-FC76250081D8 *EXPLOIT*  
| 578AA185-2908-5E4B-9A4B-3C01D25E8897 10.0 https://vulners.com/githubexploit/578AA185-2908-5E4B-9A4B-3C01D25E8897 *EXPLOIT*  
| 59C3A0A4-437D-5156-8197-A027A9A666C 10.0 https://vulners.com/githubexploit/59C3A0A4-437D-5156-8197-A027A9A666C *EXPLOIT*  
| 59588586-73CA-5907-81CA-54606591DFA4 10.0 https://vulners.com/githubexploit/59588586-73CA-5907-81CA-54606591DFA4 *EXPLOIT*  
| 500C71C1-8183-56FA-B198-1525F11832F1 10.0 https://vulners.com/githubexploit/500C71C1-8183-56FA-B198-1525F11832F1 *EXPLOIT*  
| 4D01A3A1-75C7-5730-8C5D-B2EC1F56532F 10.0 https://vulners.com/githubexploit/4D01A3A1-75C7-5730-8C5D-B2EC1F56532F *EXPLOIT*  
| 308FA41E-90C2-55C2-9E1E-B8AE02299A48 10.0 https://vulners.com/githubexploit/308FA41E-90C2-55C2-9E1E-B8AE02299A48 *EXPLOIT*  
| 344AF37C-35F5-5A70-83E4-B89587233D0C 10.0 https://vulners.com/githubexploit/344AF37C-35F5-5A70-83E4-B89587233D0C *EXPLOIT*  
| 322D01ED-0331-573E-9AAC-5868C2895F9 10.0 https://vulners.com/githubexploit/322D01ED-0331-573E-9AAC-5868C2895F9 *EXPLOIT*
```

3. Vulnerability Findings

The assessment identified multiple intentionally vulnerable services. The screenshots below preserve the observed evidence used to support the findings.

04 Vulnerability Finding 1

```
Session Actions Edit View Help
| 1337DAY-ID-26494 7.8 https://vulners.com/zdt/1337DAY-ID-26494 *EXPLOIT*
| 1021308E-F683-588B-8603-353173626207 7.8 https://vulners.com/githubexploit/1021308E-F683-588B-8603-353173626207 *EXPLOIT*
| SSV:92579 7.5 https://vulners.com/seebug/SSV:92579 *EXPLOIT*
| SSV:61450 7.5 https://vulners.com/seebug/SSV:61450 *EXPLOIT*
| EDB-ID-40963 7.5 https://vulners.com/exploitdb/EDB-ID-40963 *EXPLOIT*
| CVE-2026-60000 7.5 https://vulners.com/cve/CVE-2026-60000
| CVE-2026-59999 7.5 https://vulners.com/cve/CVE-2026-59999
| CVE-2016-18788 7.5 https://vulners.com/cve/CVE-2016-18788
| CVE-2016-10009 7.5 https://vulners.com/cve/CVE-2016-10009
| CVE-2014-1692 7.5 https://vulners.com/cve/CVE-2014-1692
| CVE-2018-5107 7.5 https://vulners.com/cve/CVE-2018-5107
| CFS2FA19-85D8-5014-850F-2411851976E2 7.5 https://vulners.com/githubexploit/CFS2FA19-85D8-5014-850F-2411851976E2 *EXPLOIT*
| 1337DAY-ID-26576 7.5 https://vulners.com/zdt/1337DAY-ID-26576 *EXPLOIT*
| SSV:92582 7.2 https://vulners.com/seebug/SSV:92582 *EXPLOIT*
| EDB-ID-40962 7.0 https://vulners.com/exploitdb/EDB-ID-40962 *EXPLOIT*
| CVE-2016-10010 7.0 https://vulners.com/cve/CVE-2016-10010
| CVE-2015-6564 7.0 https://vulners.com/cve/CVE-2015-6564
| SSV:92580 6.9 https://vulners.com/seebug/SSV:92580 *EXPLOIT*
| 1337DAY-ID-26577 6.9 https://vulners.com/zdt/1337DAY-ID-26577 *EXPLOIT*
| EDB-ID-46516 6.8 https://vulners.com/exploitdb/EDB-ID-46516 *EXPLOIT*
| EDB-ID-46193 6.8 https://vulners.com/exploitdb/EDB-ID-46193 *EXPLOIT*
| CVE-2019-6110 6.8 https://vulners.com/cve/CVE-2019-6110
| CVE-2019-6109 6.8 https://vulners.com/cve/CVE-2019-6109
| 1337DAY-ID-32328 6.8 https://vulners.com/zdt/1337DAY-ID-32328 *EXPLOIT*
| 1337DAY-ID-32009 6.8 https://vulners.com/zdt/1337DAY-ID-32009 *EXPLOIT*
| DB7C1C7-4DB6-55E8-BC08-5659FBF3AE16 6.5 https://vulners.com/githubexploit/DB7C1C7-4DB6-55E8-BC08-5659FBF3AE16 *EXPLOIT*
| D194D28F-E0D2-588B-A982-3CC562F8E00 6.5 https://vulners.com/githubexploit/D194D28F-E0D2-588B-A982-3CC562F8E00 *EXPLOIT*
| CVE-2026-60001 6.5 https://vulners.com/cve/CVE-2026-60001
| CVE-2026-59998 6.5 https://vulners.com/cve/CVE-2026-59998
| CVE-2026-35387 6.5 https://vulners.com/cve/CVE-2026-35387
| CVE-2023-51385 6.5 https://vulners.com/cve/CVE-2023-51385
| CVE-2014-2653 6.5 https://vulners.com/cve/CVE-2014-2653
| CVE-2012-0814 6.5 https://vulners.com/cve/CVE-2012-0814
| CVE-2008-1657 6.5 https://vulners.com/cve/CVE-2008-1657
| C07ADB46-2488-57B7-8375-9C761F4750A2 6.5 https://vulners.com/githubexploit/C07ADB46-2488-57B7-8375-9C761F4750A2 *EXPLOIT*
| A88CD03E-67CC-51CC-97FB-A0CACB6808C 6.5 https://vulners.com/githubexploit/A88CD03E-67CC-51CC-97FB-A0CACB6808C *EXPLOIT*
| 65815AA1-2ABD-53C1-9499-69EBA3619F1C 6.5 https://vulners.com/githubexploit/65815AA1-2ABD-53C1-9499-69EBA3619F1C *EXPLOIT*
| 5325A906-132B-590C-B0E1-6C8185252732 6.5 https://vulners.com/gitee/5325A906-132B-590C-B0E1-6C8185252732 *EXPLOIT*
| 530326CF-6AB3-5643-AA16-73DC8CB44742 6.5 https://vulners.com/githubexploit/530326CF-6AB3-5643-AA16-73DC8CB44742 *EXPLOIT*
| EDB-ID-40858 6.4 https://vulners.com/exploitdb/EDB-ID-40858 *EXPLOIT*
| EDB-ID-39569 6.4 https://vulners.com/exploitdb/EDB-ID-39569 *EXPLOIT*
| CVE-2016-3115 6.4 https://vulners.com/cve/CVE-2016-3115
| CVE-2015-6563 6.4 https://vulners.com/cve/CVE-2015-6563
| CVE-2016-10011 6.2 https://vulners.com/cve/CVE-2016-10011
| PACKETSTORM:181223 5.9 https://vulners.com/packetstorm/PACKETSTORM:181223 *EXPLOIT*
| MSF-AUXILIARY-SCANNER-SSH-ENUMUSERS- 5.9 https://vulners.com/metasploit/MSF-AUXILIARY-SCANNER-SSH-ENUMUSERS- *EXPLOIT*
| FEF0EB06-7708-5ADF-857C-1704B7AC3FE4 5.9 https://vulners.com/githubexploit/FEF0EB06-7708-5ADF-857C-1704B7AC3FE4 *EXPLOIT*
| FD20EBEA-ED84-5304-8862-84BCDEB2F288 5.9 https://vulners.com/githubexploit/FD20EBEA-ED84-5304-8862-84BCDEB2F288 *EXPLOIT*
| EDB-ID-45939 5.9 https://vulners.com/exploitdb/EDB-ID-45939 *EXPLOIT*
| EDB-ID-45233 5.9 https://vulners.com/exploitdb/EDB-ID-45233 *EXPLOIT*
| EDB-ID-45210 5.9 https://vulners.com/exploitdb/EDB-ID-45210 *EXPLOIT*
| EDB-ID-40136 5.9 https://vulners.com/exploitdb/EDB-ID-40136 *EXPLOIT*
| EDB-ID-40113 5.9 https://vulners.com/exploitdb/EDB-ID-40113 *EXPLOIT*
```

05 Vulnerability Finding 2

```
kali@kali ~  
Session Actions Edit View Help  
| 1337DAY-ID-27896 0.0 https://vulners.com/zdt/1337DAY-ID-27896 *EXPLOIT*  
80/tcp open http Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
| http-enumer:  
| /ikiwiki/: Ikiwiki  
| /test/: Test page  
| /phpinfo.php: Possible information file  
| /phpMyAdmin/: phpMyAdmin  
| /doc/: Potentially interesting directory w/ listing on 'apache/2.2.8 (ubuntu) dav/2'  
| /icons/: Potentially interesting folder w/ directory listing  
| /index/: Potentially interesting folder  
| http-csrf:  
| Spidering limited to: maxdepth=3; maxpagecount=20; withinhost=192.168.56.102  
| Found the following possible CSRF vulnerabilities:  
| Path: http://192.168.56.102:80/dvwa/  
| Form id:  
| Form action: login.php  
| Path: http://192.168.56.102:80/twiki/TWikiDocumentation.html  
| Form id:  
| Form action: http://Twiki.org/cgi-bin/passwd/TWiki/WebHome  
| Path: http://192.168.56.102:80/twiki/TWikiDocumentation.html  
| Form id:  
| Form action: http://Twiki.org/cgi-bin/passwd/Main/WebHome  
| Path: http://192.168.56.102:80/twiki/TWikiDocumentation.html  
| Form id:  
| Form action: http://Twiki.org/cgi-bin/edit/TWiki/  
| Path: http://192.168.56.102:80/twiki/TWikiDocumentation.html  
| Form id:  
| Form action: http://Twiki.org/cgi-bin/view/TWiki/TWikiSkins  
| Path: http://192.168.56.102:80/twiki/TWikiDocumentation.html  
| Form id:  
| Form action: http://Twiki.org/cgi-bin/manage/TWiki/ManagingWebs  
| Path: http://192.168.56.102:80/dvwa/login.php  
| Form id:  
| Form action: login.php  
| http-fileupload-exploiter:  
| Couldn't find a file-type field.  
| http-vuln-cve2017-1001000: ERROR: Script execution failed (use -d to debug)  
| http-stored-xss: Couldn't find any stored XSS vulnerabilities.  
| vulners:  
| cpe:/a:apache:http_server:2.2.8:  
| SSV:69341 10.0 https://vulners.com/seebug/SSV:69341 *EXPLOIT*  
| SSV:19282 10.0 https://vulners.com/seebug/SSV:19282 *EXPLOIT*  
| SSV:19236 10.0 https://vulners.com/seebug/SSV:19236 *EXPLOIT*  
| SSV:11999 10.0 https://vulners.com/seebug/SSV:11999 *EXPLOIT*  
| PACKETSTORM:86964 10.0 https://vulners.com/packetstorm/PACKETSTORM:86964 *EXPLOIT*
```

06 Vulnerability Finding 3

```
kali@kali ~  
Session Actions Edit View Help  
| CVE-2007-5925 4.0 https://vulners.com/cve/CVE-2007-5925  
| CVE-2010-1626 3.6 https://vulners.com/cve/CVE-2010-1626  
| 5832/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7  
| ssl-ccs-injection:  
| VULNERABLE:  
| SSL/TLS MITM vulnerability (CCS Injection)  
| State: VULNERABLE  
| Risk factor: High  
| OpenSSL before 0.9.8za, 1.0.0 before 1.0.0m, and 1.0.1 before 1.0.1h  
| does not properly restrict processing of ChangeCipherSpec messages,  
| which allows man-in-the-middle attackers to trigger use of a zero  
| length master key in certain OpenSSL-to-OpenSSL communications, and  
| consequently hijack sessions or obtain sensitive information, via  
| a crafted TLS handshake, aka the "CCS Injection" vulnerability.  
|  
| References:  
| http://www.cvedetails.com/cve/2014-0224  
| http://www.openssl.org/news/secadv_20140605.txt  
| https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2014-0224  
|  
| ssl-poodle:  
| VULNERABLE:  
| SSL POODLE information leak  
| State: VULNERABLE  
| IDs: BID:70574 CVE:CVE-2014-3566  
| The SSL protocol 3.0, as used in OpenSSL through 1.0.1i and other  
| products, uses nondeterministic CBC padding, which makes it easier  
| for man-in-the-middle attackers to obtain cleartext data via a  
| padding-oracle attack, aka the "POODLE" issue.  
| Disclosure date: 2014-10-14  
| Check results:  
| TLS_RSA_WITH_AES_128_CBC_SHA  
| References:  
| https://www.openssl.org/~bodo/ssl-poodle.pdf  
| https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2014-3566  
| https://www.imperialviolet.org/2014/10/14/poodle.html  
| https://www.securityfocus.com/bid/70574  
|  
| vulners:  
| cpe:/a:postgresql:postgresql:8.3:  
| SSV:60718 10.0 https://vulners.com/seebug/SSV:60718 *EXPLOIT*  
| CVE-2013-1903 10.0 https://vulners.com/cve/CVE-2013-1903  
| CVE-2013-1902 10.0 https://vulners.com/cve/CVE-2013-1902  
| PACKETSTORM:216210 9.9 https://vulners.com/packetstorm/PACKETSTORM:216210 *EXPLOIT*  
| POSTGRESQL:CVE-2019-10211 9.8 https://vulners.com/postgresql/POSTGRESQL:CVE-2019-10211  
| POSTGRESQL:CVE-2018-16850 9.8 https://vulners.com/postgresql/POSTGRESQL:CVE-2018-16850  
| POSTGRESQL:CVE-2017-7546 9.8 https://vulners.com/postgresql/POSTGRESQL:CVE-2017-7546  
| POSTGRESQL:CVE-2015-3166 9.8 https://vulners.com/postgresql/POSTGRESQL:CVE-2015-3166  
| POSTGRESQL:CVE-2015-0244 9.8 https://vulners.com/postgresql/POSTGRESQL:CVE-2015-0244  
| PACKETSTORM:189316 9.8 https://vulners.com/packetstorm/PACKETSTORM:189316 *EXPLOIT*  
| CVE-2019-10211 9.8 https://vulners.com/cve/CVE-2019-10211  
| CVE-2015-3166 9.8 https://vulners.com/cve/CVE-2015-3166  
| CVE-2015-0244 9.8 https://vulners.com/cve/CVE-2015-0244  
| CNVD-2020-02196 9.8 https://vulners.com/cnvd/CNVD-2020-02196  
| CNVD-2017-26577 9.8 https://vulners.com/cnvd/CNVD-2017-26577
```

07 Vulnerability Finding 4

```
kali@kali ~  
Session Actions Edit View Help  
| CVE-2014-0060 4.0 https://vulners.com/cve/CVE-2014-0060  
| CVE-2012-2655 4.0 https://vulners.com/cve/CVE-2012-2655  
| CVE-2009-3229 4.0 https://vulners.com/cve/CVE-2009-3229  
| PostgreSQL-CVE-2026-6469 3.8 https://vulners.com/postgresql/PostgreSQL-CVE-2026-6469  
| PostgreSQL-CVE-2026-16241 3.8 https://vulners.com/postgresql/PostgreSQL-CVE-2026-16241  
| PostgreSQL-CVE-2026-14673 3.8 https://vulners.com/postgresql/PostgreSQL-CVE-2026-14673  
| PostgreSQL-CVE-2024-10977 3.7 https://vulners.com/postgresql/PostgreSQL-CVE-2024-10977  
| PostgreSQL-CVE-2022-41862 3.7 https://vulners.com/postgresql/PostgreSQL-CVE-2022-41862  
| SSV-19322 3.5 https://vulners.com/seebug/SSV-19322 *EXPLOIT*  
| PostgreSQL-CVE-2019-10209 3.5 https://vulners.com/postgresql/PostgreSQL-CVE-2019-10209  
| PACKETSTORM-127892 3.5 https://vulners.com/packetstorm/PACKETSTORM-127892 *EXPLOIT*  
| EDB-ID-33729 3.5 https://vulners.com/exploitdb/EDB-ID-33729 *EXPLOIT*  
| CVE-2018-0733 3.5 https://vulners.com/cve/CVE-2018-0733  
| PostgreSQL-CVE-2025-8713 3.1 https://vulners.com/postgresql/PostgreSQL-CVE-2025-8713  
| PostgreSQL-CVE-2025-12817 3.1 https://vulners.com/postgresql/PostgreSQL-CVE-2025-12817  
| ssl-dh-params:  
| VULNERABLE:  
| Diffie-Hellman Key Exchange Insufficient Group Strength  
| State: VULNERABLE  
| Transport Layer Security (TLS) services that use Diffie-Hellman groups  
| of insufficient strength, especially those using one of a few commonly  
| shared groups, may be susceptible to passive eavesdropping attacks.  
| Check results:  
| WEAK DH GROUP 1  
| Cipher Suite: TLS_DHE_RSA_WITH_3DES_EDE_CBC_SHA  
| Modulus Type: Safe prime  
| Modulus Source: Unknown/Custom-generated  
| Modulus Length: 1024  
| Generator Length: 8  
| Public Key Length: 1024  
| References:  
| https://weakdh.org  
| 5980/tcp open vnc VNC (protocol 3.3)  
| 6080/tcp open X11 (access denied)  
| 6667/tcp open irc UnrealIRCd  
| _irc-unrealircd-backdoor: Looks like trojaned version of unrealircd. See http://seclists.org/fulldisclosure/2010/Jun/277  
| 8080/tcp open ajp13 Apache Jserv (Protocol v1.3)  
| 8180/tcp open http Apache Tomcat/Coyote JSP engine 1.1  
| _http-stored-xss: Couldn't find any stored XSS vulnerabilities.  
| _http-server-header: Apache-Coyote/1.1  
| _http-csrf: Couldn't find any CSRF vulnerabilities.  
| _http-dombased-xss: Couldn't find any DOM based XSS.  
| MAC Address: 08:00:27:02:32:8E (Oracle VirtualBox virtual NIC)  
| Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel  
Host script results:  
|_ smb-vuln-ms10-054: false  
|_ smb-vuln-regsvc-dos: ERROR: Script execution failed (use -d to debug)  
|_ smb-vuln-ms10-061: false  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 364.19 seconds  
msf > |
```

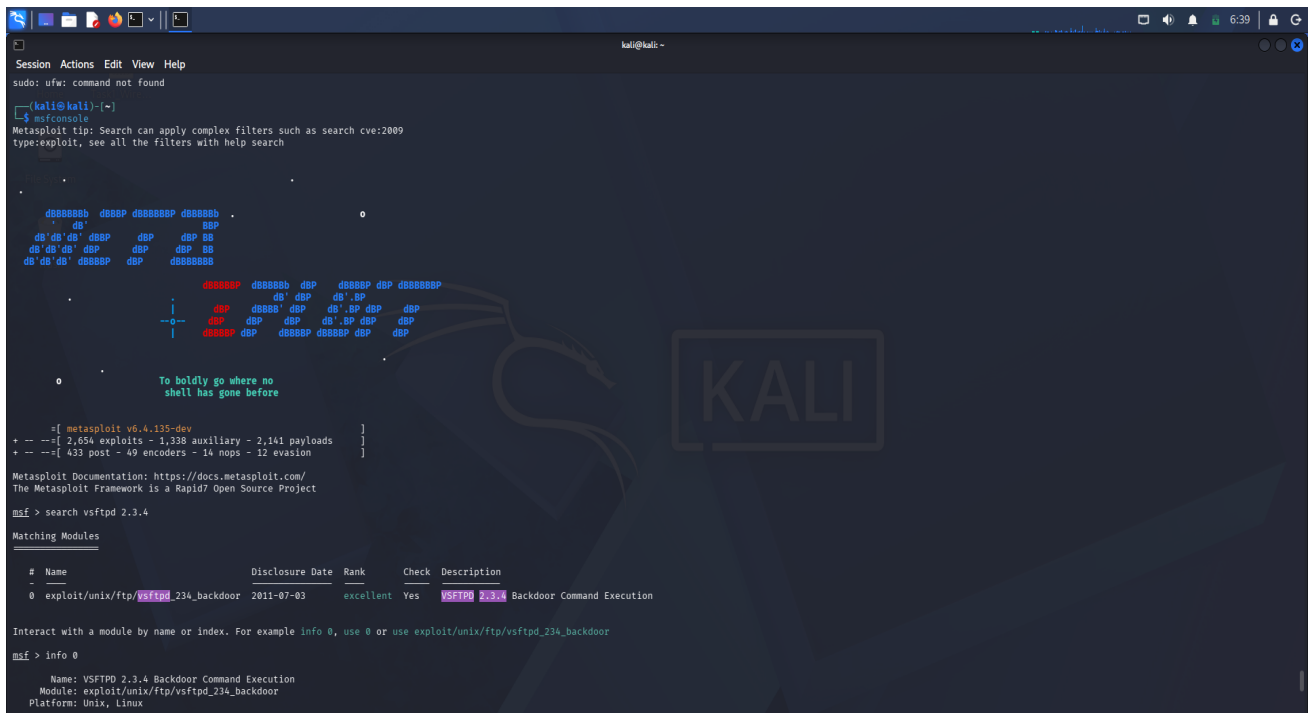
4. VSFTPD 2.3.4 — Vulnerability Validation

TCP/21 was identified as running vsftpd 2.3.4. Metasploit was opened and the VSFTPD 2.3.4 backdoor module was selected. Module information identifies the historical backdoor and CVE-2011-2523. The module options show the target host and FTP port as required parameters.

Impact: A vulnerable/backdoored FTP service can enable unauthorized command execution and compromise of the target.

Mitigation: Remove or upgrade the vulnerable service, restrict FTP exposure, prefer secure file transfer, and monitor logs.

Metasploit VSFTPD 2.3.4 module



```
kali@kali: ~  
Session Actions Edit View Help  
sudo: ufw: command not found  
kali@kali:~$ msfconsole  
Metasploit tip: Search can apply complex filters such as search cve:2009  
type:exploit, see all the filters with help search  
msf> search vsftpd 2.3.4  
Matching Modules  
# Name Disclosure Date Rank Check Description  
0 exploit/unix/ftp/vsftpd_234_backdoor 2011-07-03 excellent Yes VSFTPD 2.3.4 Backdoor Command Execution  
Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/ftp/vsftpd_234_backdoor  
msf> info 0  
Name: VSFTPD 2.3.4 Backdoor Command Execution  
Module: exploit/unix/ftp/vsftpd_234_backdoor  
Platform: Unix, Linux
```

Module information and CVE reference

```
kali@kali: ~  
Session Actions Edit View Help  
# Name Disclosure Date Rank Check Description  
0 exploit/unix/ftp/vsftpd_234_backdoor 2011-07-03 excellent Yes VSFTPD 2.3.4 Backdoor Command Execution  
  
Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/ftp/vsftpd_234_backdoor  
msf > info 0  
  
Name: VSFTPD 2.3.4 Backdoor Command Execution  
Module: exploit/unix/ftp/vsftpd_234_backdoor  
Platform: Unix, Linux  
Arch: cmd  
Privileged: Yes  
License: Metasploit Framework License (BSD)  
Rank: Excellent  
Disclosed: 2011-07-03  
  
Provided by:  
hdm <x@hdm.io>  
MC <mc@metasploit.com>  
g0tmilk  
  
Module side effects:  
unknown-side-effects  
  
Module stability:  
unknown-stability  
  
Module reliability:  
unknown-reliability  
  
Available targets:  
Id Name  
--  
0 Linux/Unix Command  
  
Check supported:  
Yes  
  
Basic options:  
Name Current Setting Required Description  
-----  
RHOSTS 21 yes The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html  
RPORT 21 yes The target port (TCP)  
  
Payload information:  
Space: 2000  
Avoid: 0 characters  
  
Description:  
This module exploits a malicious backdoor that was added to the VSFTPD download archive. This backdoor was introduced into the vsftpd-2.3.4.tar.gz archive between
```

Module configuration options

```
msf > use 0
[*] Using configured payload cmd/linux/http/x86/meterpreter_reverse_tcp
msf exploit(winx/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/winx/ftp/vsftpd_234_backdoor):

Name      Current Setting  Required  Description
--      -
RHOSTS    21               yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT     21               yes       The target port (TCP)

Payload options (cmd/linux/http/x86/meterpreter_reverse_tcp):

Name      Current Setting  Required  Description
--      -
FETCH_COMMAND  CURL            yes       Command to fetch payload (Accepted: CURL, FTP, TFTP, WGET)
FETCH_DELETE   false           yes       Attempt to delete the binary after execution
FETCH_FILELESS none            yes       Attempt to run payload without touching disk by using anonymous handles, requires Linux >= 3.17 (for Python variant also Python >= 3.8, tested shells are sh, bash, zsh) (Accepted: none, python)
FETCH_SRVHOST  no              no        Local IP to use for serving payload
FETCH_SRVPORT  8080            yes       Local port to use for serving payload
FETCH_URI_PATH no              no        Local URI to use for serving payload
LHOST        yes             yes       The listen address (an interface may be specified)
LPORT        4444            yes       The listen port

When FETCH_COMMAND is one of CURL,GET,WGET:

Name      Current Setting  Required  Description
--      -
FETCH_PIPE false           yes       Host both the binary payload and the command so it can be piped directly to the shell.

When FETCH_FILELESS is none:

Name      Current Setting  Required  Description
--      -
FETCH_FILENAME bddMJqDEaFck    no        Name to use on remote system when storing payload; cannot contain spaces or slashes
FETCH_WRITABLE_DIR ./               yes       Remote writable dir to store payload; cannot contain spaces

Exploit target:

Id  Name
--  -
0   Linux/Unix Command

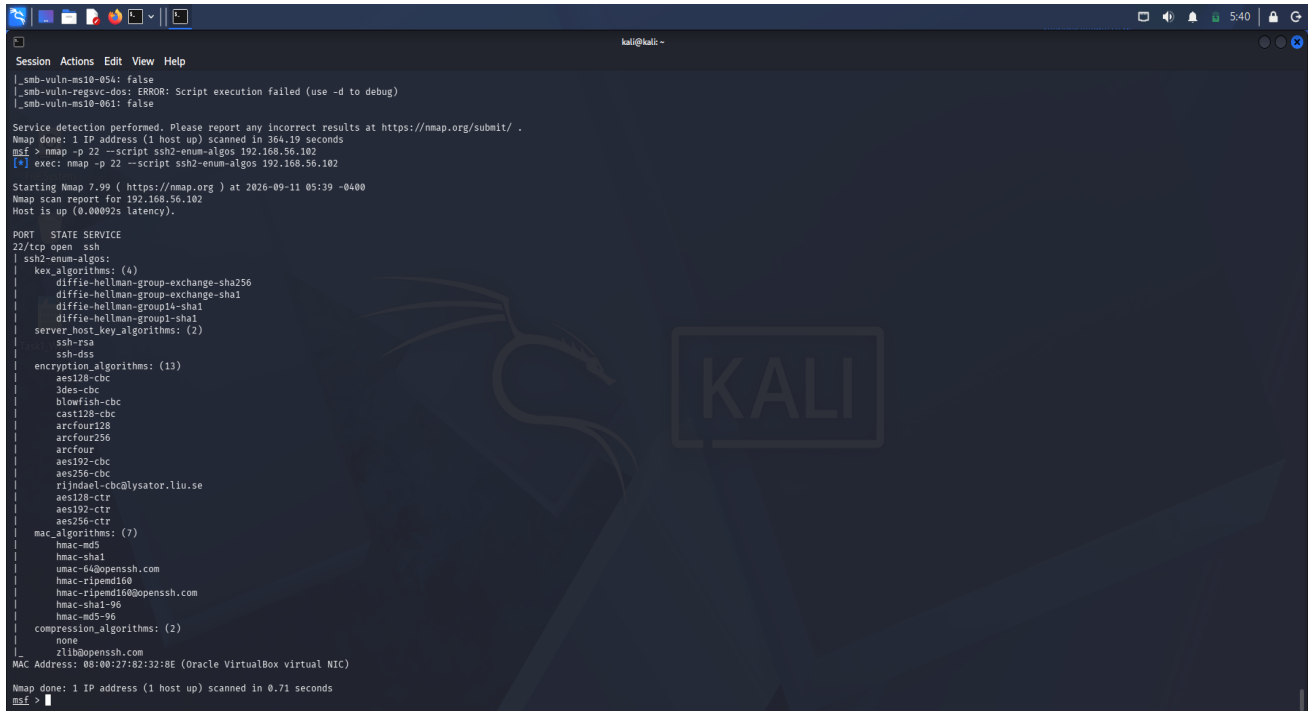
View the full module info with the info, or info -d command.
msf exploit(winx/ftp/vsftpd_234_backdoor) >
```

5. SSH Security Assessment

SSH was observed on TCP/22. Algorithm enumeration showed legacy options, including SHA-1 based key-exchange algorithms, SSH-RSA/SSH-DSS host-key algorithms, older CBC/ARCFOUR-style encryption options, and legacy MACs. These observations support a recommendation to modernize the SSH configuration.

Mitigation: Disable deprecated algorithms, use a current OpenSSH release, restrict SSH to trusted networks, and use strong authentication controls.

SSH security assessment



```
Session Actions Edit View Help
|_smb-vuln-ms10-054: false
|_smb-vuln-regsvc-dos: ERROR: Script execution failed (use -d to debug)
|_smb-vuln-ms10-061: false

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 364.19 seconds
msf > nmap -p 22 --script ssh2-enum-algos 192.168.56.102
[*] exec: nmap -p 22 --script ssh2-enum-algos 192.168.56.102

Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-11 05:39 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0002s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
|_ssh2-enum-algos:
|_  kex_algorithms: (4)
|_    diffie-hellman-group-exchange-sha256
|_    diffie-hellman-group-exchange-sha1
|_    diffie-hellman-group14-sha1
|_    diffie-hellman-group1-sha1
|_  server_host_key_algorithms: (2)
|_    ssh-rsa
|_    ssh-dss
|_  encryption_algorithms: (13)
|_    aes128-cbc
|_    3des-cbc
|_    blowfish-cbc
|_    cast128-cbc
|_    arcfour128
|_    arcfour256
|_    arcfour
|_    aes192-cbc
|_    aes256-cbc
|_    rijndael-cbc@lysator.liu.se
|_    aes128-ctr
|_    aes192-ctr
|_    aes256-ctr
|_  mac_algorithms: (7)
|_    hmac-md5
|_    hmac-sha1
|_    umac-64@openssh.com
|_    hmac-ripemd160
|_    hmac-ripemd160@openssh.com
|_    hmac-sha1-96
|_    hmac-md5-96
|_  compression_algorithms: (2)
|_    none
|_    zlib@openssh.com
|_  MAC Address: 08:00:27:182:32:8E (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.71 seconds
msf >
```

SSH algorithm enumeration

```

kali@kali: ~
Session Actions Edit View Help
0 Linux/Unix Command

View the full module info with the info, or info -d command.

msf exploit(unix/ftp/vsftpd_234_backdoor) > nmap -p 22 --script ssh2-enum-algos 192.168.56.102
[*] exec: nmap -p 22 --script ssh2-enum-algos 192.168.56.102

Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-11 06:45 -0400
Nmap scan report for 192.168.56.102
Host is up (0.012s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
| ssh2-enum-algos:
|   key_algorithms: (4)
|     diffie-hellman-group-exchange-sha256
|     diffie-hellman-group-exchange-sha1
|     diffie-hellman-group14-sha1
|     diffie-hellman-group1-sha1
|   server_host_key_algorithms: (2)
|     ssh-rsa
|     ssh-dss
|   encryption_algorithms: (13)
|     aes128-cbc
|     3des-cbc
|     blowfish-cbc
|     cast128-cbc
|     arcfour128
|     arcfour256
|     arcfour
|     aes192-cbc
|     aes256-cbc
|     rijndael-cbc@lysator.liu.se
|     aes128-ctr
|     aes192-ctr
|     aes256-ctr
|   mac_algorithms: (7)
|     hmac-md5
|     hmac-sha1
|     umac-64@openssh.com
|     hmac-ripemd160
|     hmac-ripemd160@openssh.com
|     hmac-sha1-96
|     hmac-md5-96
|   compression_algorithms: (2)
|     none
|     zlib@openssh.com
MAC Address: 08:00:27:62:32:8E (Oracle VirtualBox virtual NIC)

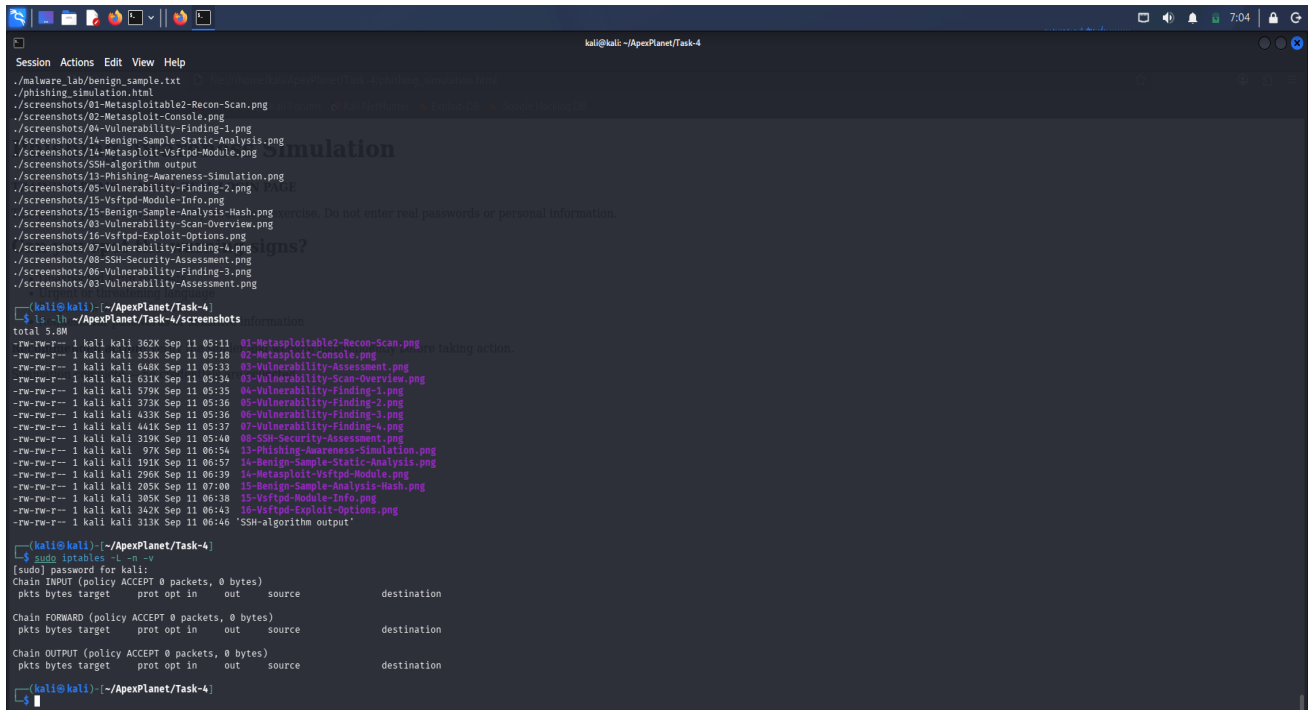
nmap done: 1 IP address (1 host up) scanned in 0.77 seconds
msf exploit(unix/ftp/vsftpd_234_backdoor) >

```

6. System Hardening Assessment

The lab included evidence collection for firewall state, listening services, and patch/upgrade status. The recommended hardening actions are to restrict inbound traffic, disable unnecessary services, and apply security updates.

Firewall state



```
kali@kali: ~/ApexPlanet/Task-4
Session Actions Edit View Help
./malware_lab/benign_sample.txt
./phishing_simulation.html
./screenshots/01-Metasploitable2-Recon-Scan.png
./screenshots/02-Metasploit-Console.png
./screenshots/04-Vulnerability-Finding-1.png
./screenshots/14-Benign-Sample-Static-Analysis.png
./screenshots/14-Metasploit-Vsftpd-Module.png
./screenshots/SSH-algorithm_output
./screenshots/13-Phishing-Awareness-Simulation.png
./screenshots/05-Vulnerability-Finding-2.png
./screenshots/15-Vsftpd-Module-Info.png
./screenshots/15-Benign-Sample-Analysis-Hash.png
./screenshots/03-Vulnerability-Scan-Overview.png
./screenshots/16-Vsftpd-Exploit-Options.png
./screenshots/07-Vulnerability-Finding-4.png
./screenshots/08-SSH-Security-Assessment.png
./screenshots/06-Vulnerability-Finding-3.png
./screenshots/03-Vulnerability-Assessment.png

kali@kali:~/ApexPlanet/Task-4$ ls -lh ~/ApexPlanet/Task-4/screenshots
total 5.8M
-rw-rw-r-- 1 kali kali 362K Sep 11 05:11 01-Metasploitable2-Recon-Scan.png
-rw-rw-r-- 1 kali kali 352K Sep 11 05:18 02-Metasploit-Console.png
-rw-rw-r-- 1 kali kali 648K Sep 11 05:33 03-Vulnerability-Assessment.png
-rw-rw-r-- 1 kali kali 631K Sep 11 05:34 03-Vulnerability-Scan-Overview.png
-rw-rw-r-- 1 kali kali 579K Sep 11 05:35 04-Vulnerability-Finding-1.png
-rw-rw-r-- 1 kali kali 372K Sep 11 05:36 05-Vulnerability-Finding-2.png
-rw-rw-r-- 1 kali kali 433K Sep 11 05:36 06-Vulnerability-Finding-3.png
-rw-rw-r-- 1 kali kali 441K Sep 11 05:37 07-Vulnerability-Finding-4.png
-rw-rw-r-- 1 kali kali 319K Sep 11 05:40 08-SSH-Security-Assessment.png
-rw-rw-r-- 1 kali kali 97K Sep 11 06:54 13-Phishing-Awareness-Simulation.png
-rw-rw-r-- 1 kali kali 191K Sep 11 06:57 14-Benign-Sample-Static-Analysis.png
-rw-rw-r-- 1 kali kali 296K Sep 11 06:39 14-Metasploit-Vsftpd-Module.png
-rw-rw-r-- 1 kali kali 285K Sep 11 07:00 15-Benign-Sample-Analysis-Hash.png
-rw-rw-r-- 1 kali kali 288K Sep 11 06:38 15-Vsftpd-Module-Info.png
-rw-rw-r-- 1 kali kali 342K Sep 11 06:43 16-Vsftpd-Exploit-Options.png
-rw-rw-r-- 1 kali kali 313K Sep 11 06:46 'SSH-algorithm output'

kali@kali:~/ApexPlanet/Task-4$ sudo iptables -t -n -v
[sudo] password for kali:
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in      out     source      destination

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in      out     source      destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in      out     source      destination
```

Listening services

```
kali@kali: ~/ApexPlanet/Task-4

Session Actions Edit View Help

./screenshots/SSH-algorithm-output.png
./screenshots/13-Phishing-Awareness-Simulation.png
./screenshots/05-Vulnerability-Finding-2.png
./screenshots/15-Vsftpd-Module-Info.png
./screenshots/15-Benign-Sample-Analysis-Hash.png
./screenshots/03-Vulnerability-Scan-Overview.png
./screenshots/16-Vsftpd-Exploit-Options.png
./screenshots/07-Vulnerability-Finding-4.png
./screenshots/08-SSH-Security-Assessment.png
./screenshots/06-Vulnerability-Finding-3.png
./screenshots/03-Vulnerability-Assessment.png

Simulation

Business exercise. Do not enter real passwords or personal information.

kali@kali:~/ApexPlanet/Task-4$ ls -lh ~/ApexPlanet/Task-4/screenshots
total 5.8M
-rw-rw-r-- 1 kali kali 362K Sep 11 05:11 01-Metasploitable2-Recon-Scan.png
-rw-rw-r-- 1 kali kali 353K Sep 11 05:18 02-Metasploit-Console.png
-rw-rw-r-- 1 kali kali 648K Sep 11 05:33 03-Vulnerability-Assessment.png
-rw-rw-r-- 1 kali kali 631K Sep 11 05:34 03-Vulnerability-Scan-Overview.png
-rw-rw-r-- 1 kali kali 579K Sep 11 05:35 04-Vulnerability-Finding-1.png
-rw-rw-r-- 1 kali kali 373K Sep 11 05:36 05-Vulnerability-Finding-2.png
-rw-rw-r-- 1 kali kali 433K Sep 11 05:36 06-Vulnerability-Finding-3.png
-rw-rw-r-- 1 kali kali 441K Sep 11 05:37 07-Vulnerability-Finding-4.png
-rw-rw-r-- 1 kali kali 319K Sep 11 05:40 08-SSH-Security-Assessment.png
-rw-rw-r-- 1 kali kali 97K Sep 11 06:54 13-Phishing-Awareness-Simulation.png
-rw-rw-r-- 1 kali kali 191K Sep 11 06:57 14-Benign-Sample-Static-Analysis.png
-rw-rw-r-- 1 kali kali 296K Sep 11 06:59 14-Metasploit-Vsftpd-Module.png
-rw-rw-r-- 1 kali kali 285K Sep 11 07:00 15-Benign-Sample-Analysis-Hash.png
-rw-rw-r-- 1 kali kali 385K Sep 11 06:38 15-Vsftpd-Module-Info.png
-rw-rw-r-- 1 kali kali 342K Sep 11 06:43 16-Vsftpd-Exploit-Options.png
-rw-rw-r-- 1 kali kali 313K Sep 11 06:46 SSH-algorithm-output.png

kali@kali:~/ApexPlanet/Task-4$ sudo iptables -t n -v
[sudo] password for kali:
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in      out     source            destination

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in      out     source            destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in      out     source            destination

kali@kali:~/ApexPlanet/Task-4$ netstat -tlnm
(Not all processes could be identified, non-owned process info
will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
$
```

Patch assessment

```
kali@kali: ~/ApexPlanet/Task-4

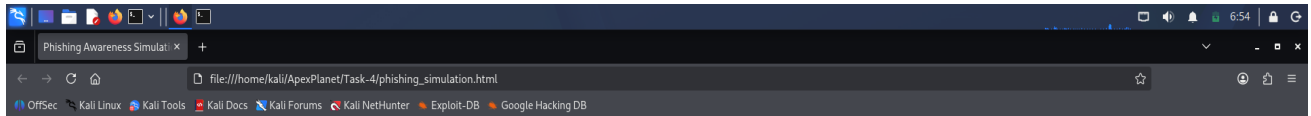
Session Actions Edit View Help

Conf ruby-tzinfo (2.0.6-2 kali-rolling [all])
Conf ruby-i18n (1.15.2-1 kali-rolling [all])
Conf ruby-connection-pool (3.0.2-2 kali-rolling [all])
Conf ruby-minitest (6.0.6-2 kali-rolling [all])
Conf ruby-activesupport (2:7.2.3.2+dfsg-2 kali-rolling [all])
Conf ruby-mini-portile2 (2.8.9-2 kali-rolling [all])
Conf ruby-nokogiri (1:19.4+dfsg-1 kali-rolling [amd64])
Conf ruby-opt-parse-validator (1.10.1-0kali2 kali-rolling [all])
Conf ruby-yajl (1.4.3-2 kali-rolling [amd64])
Conf ruby-webrick (1.9.2-2 kali-rolling [all])
Conf ruby-ssl-scanner (0.15.0-0kali24 kali-rolling [all])
Conf ruby-public-suffix (7.0.5-1 kali-rolling [all]) Do not enter real passwords or personal information.
Conf ruby-addressable (2.9.0-1 kali-rolling [all])
Conf ruby-ipaddress (0.8.3-6 kali-rolling [all])
Conf ruby-Levenshtein (0.2.2-4 kali-rolling [amd64])
Conf ruby-little-plugger (1.1.4-3 kali-rolling [all])
Conf ruby-mime-types-data (3.2026.0701-1 kali-rolling [all])
Conf ruby-net-http-digest-auth (1.4.1-2 kali-rolling [all])
Conf ruby-nori (2.9.1-1 kali-rolling [all])
Conf ruby-oj (3.17.4-1 kali-rolling [amd64])
Conf ruby-rchardet (1.10.2-1 kali-rolling [all])
Conf ruby-spider (0.8.0-2 kali-rolling [all])
Conf ruby-sqlite3 (2.9.5-2 kali-rolling [amd64]) website independently before taking action.
Conf unzip (6.0-30 kali-rolling [amd64])
Conf ruby-zip (3.4.1+dfsg-1 kali-rolling [all])
Conf samba-common (2:4.24.5+dfsg-1 kali-rolling [all])
Conf tdb-tools (2:1.4.15+samba-24.5+dfsg-1 kali-rolling [amd64])
Conf tex-gyre (20180621-8 kali-rolling [all])
Conf texlive-plain-generic (2026.20260711-2 kali-rolling [all])
Conf texlive-pictures (2026.20260711-2 kali-rolling [all])
Conf texlive-latex-base (2026.20260711-2 kali-rolling [all])
Conf texlive-latex-extra (2026.20260711-2 kali-rolling [all])
Conf texlive-latex-recommended (2026.20260711-2 kali-rolling [all])
Conf texlive-base (2026.20260711-2 kali-rolling [all])
Conf texlive-fonts-recommended (2026.20260711-2 kali-rolling [all])
Conf tpm-udev (4.1.3-7 kali-rolling [all])
Conf tumble (4.20.2-1 kali-rolling [amd64])
Conf tumble-common (4.20.2-1 kali-rolling [all])
Conf usb-modeswitch (2.6.2-2 kali-rolling [amd64])
Conf usb.ids (2026.06.26-1 kali-rolling [all])
Conf virtualbox-guest-x11 (7.2.14+dfsg-1 kali-rolling [amd64])
Conf virtualbox-guest-utils (7.2.14+dfsg-1 kali-rolling [amd64])
Conf weasyprint (69.0-2 kali-rolling [all])
Conf xbrlapi (6.9.1+repack-1 kali-rolling [amd64])
Conf xdg-desktop-portal (1.22.1+ds-1 kali-rolling [amd64])
Conf xfce4-helpers (4.20.5-1 kali-rolling [amd64])
Conf xfce4-panel (4.20.8-1 kali-rolling [amd64])
Conf xfce4-settings (4.20.5-1 kali-rolling [amd64])
Conf xserver-xephyr (2:21.1.23-1 kali-rolling [amd64])
Conf xvf (2:21.1.23-1 kali-rolling [amd64])

kali@kali:~/ApexPlanet/Task-4$
```

7. Social Engineering — Awareness Simulation

A local training page was created as an awareness-only phishing simulation. It is explicitly labelled as a training demo and does not collect, store, or transmit credentials. The page presents common phishing warning signs and recommended user actions.



Phishing Awareness Simulation

TRAINING DEMO — NOT A REAL LOGIN PAGE

This page is part of a cybersecurity awareness exercise. Do not enter real passwords or personal information.

Can you spot the warning signs?

- Unexpected login requests
- Urgent or threatening language
- Suspicious links or domains
- Requests for passwords or sensitive information

Recommended action: Verify the sender and website independently before taking action.

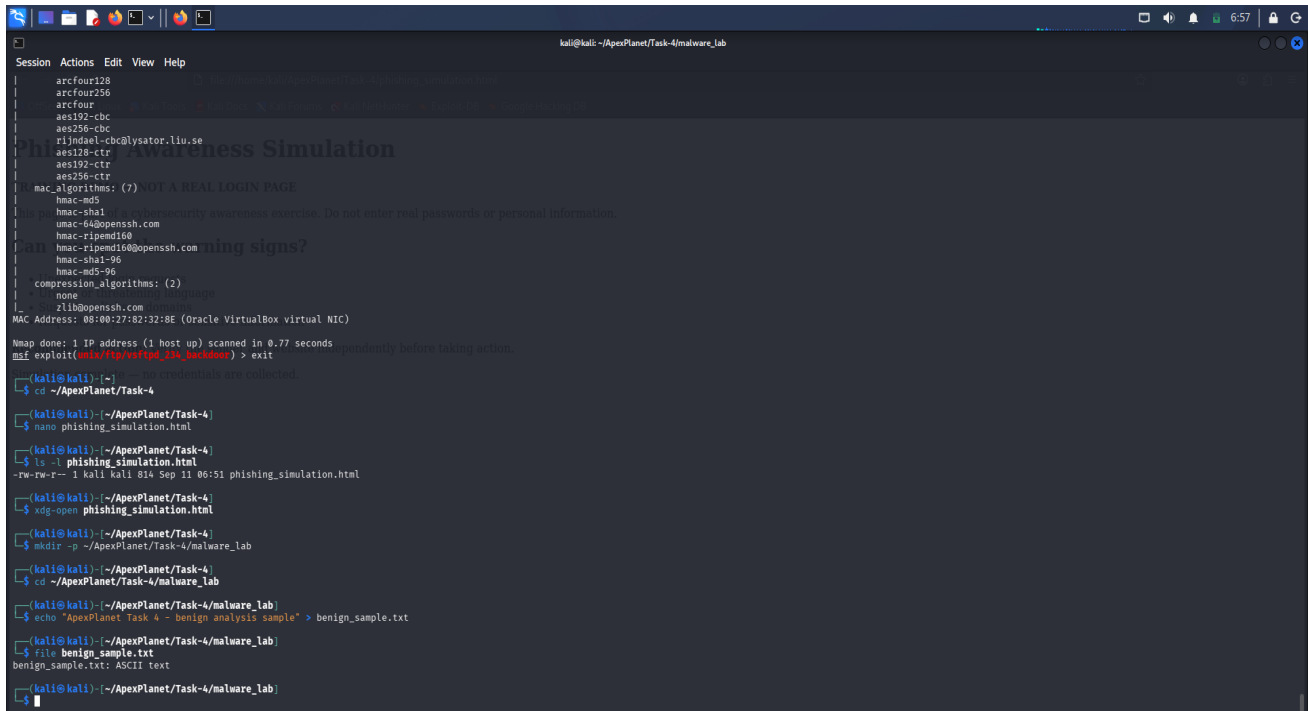
Simulation complete — no credentials are collected.

8. Malware Basics

A harmless text sample was used to demonstrate basic static analysis. The file was identified as ASCII text and a SHA-256 hash was recorded. Static analysis examines artifacts without execution; dynamic analysis observes runtime behavior in an isolated environment. No malware was executed in this exercise.

Recorded SHA-256: fbe0bc72ac2e53e449c6ef65f1f6c2acc9b3fe16e1758f33d25a8535fec7f6d6

Benign sample file-type check



```
kali@kali: ~/ApexPlanet/Task-4/malware_lab
Session Actions Edit View Help
|-----|
| arcfour128 |
| arcfour256 |
| arcfour    |
| aes192-cbc |
| aes256-cbc |
| rijndael-cbc@lysator.liu.se |
| aes128-ctr |
| aes192-ctr |
| aes256-ctr |
| mac_algorithm: (7) 0T A REAL LOGIN PAGE |
| hmac-md5   |
| hmac-sha1  |
| umac-64@openssh.com |
| hmac-ripemd160 |
| hmac-ripemd160@openssh.com |
| hmac-sha1-96 |
| hmac-md5-96 |
| compression_algorithms: (2) |
| none         |
| zlib@openssh.com |
| MAC Address: 08:00:27:82:32:8E (Oracle VirtualBox virtual NIC) |
|-----|
Nmap done: 1 IP address (1 host up) scanned in 0.77 seconds
msf exploit(multi/ftp_vsftpd_23k_backdoor) > exit
(kali@kali)-[~]
$ cd ~/ApexPlanet/Task-4
(kali@kali)-[~/ApexPlanet/Task-4]
$ nano phishing_simulation.html
(kali@kali)-[~/ApexPlanet/Task-4]
$ ls -l phishing_simulation.html
-rw-rw-r-- 1 kali kali 814 Sep 11 06:51 phishing_simulation.html
(kali@kali)-[~/ApexPlanet/Task-4]
$ xdg-open phishing_simulation.html
(kali@kali)-[~/ApexPlanet/Task-4]
$ mkdir -p ~/ApexPlanet/Task-4/malware_lab
(kali@kali)-[~/ApexPlanet/Task-4]
$ cd ~/ApexPlanet/Task-4/malware_lab
(kali@kali)-[~/ApexPlanet/Task-4/malware_lab]
$ echo "ApexPlanet Task 4 - benign analysis sample" > benign_sample.txt
(kali@kali)-[~/ApexPlanet/Task-4/malware_lab]
$ file benign_sample.txt
benign_sample.txt: ASCII text
(kali@kali)-[~/ApexPlanet/Task-4/malware_lab]
```

Benign sample content and SHA-256 hash

```
kali@kali: ~/ApexPlanet/Task-4/malware_lab
Session Actions Edit View Help
| aes256-ctr
| mac_algorithms: (7)
| hmac-md5
| hmac-sha1
| umac-64@openssh.com
| hmac-ripemd160
| hmac-ripemd160@openssh.com
| hmac-sha1-96
| hmac-md5-96
| compression_algorithms: (2)
| none
| zlib@openssh.com
MAC Address: 08:00:27:82:32:8E (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.77 seconds
msf exploit(multi/tip/sslpad_ssl_backend) > exit

kali@kali:~$ cd ~/ApexPlanet/Task-4
kali@kali:~/ApexPlanet/Task-4$ nano phishing_simulation.html
kali@kali:~/ApexPlanet/Task-4$ nano phishing_simulation.html
kali@kali:~/ApexPlanet/Task-4$ ls -l phishing_simulation.html
-rw-rw-r-- 1 kali kali 814 Sep 11 06:51 phishing_simulation.html
kali@kali:~/ApexPlanet/Task-4$ xdg-open phishing_simulation.html
kali@kali:~/ApexPlanet/Task-4$ mkdir -p ~/ApexPlanet/Task-4/malware_lab
kali@kali:~/ApexPlanet/Task-4$ cd ~/ApexPlanet/Task-4/malware_lab
kali@kali:~/ApexPlanet/Task-4/malware_lab$ echo "ApexPlanet Task 4 - Benign analysis sample" > benign_sample.txt
kali@kali:~/ApexPlanet/Task-4/malware_lab$ file benign_sample.txt
benign_sample.txt: ASCII text
kali@kali:~/ApexPlanet/Task-4/malware_lab$ cat benign_sample.txt
ApexPlanet Task 4 - Benign analysis sample
kali@kali:~/ApexPlanet/Task-4/malware_lab$ sha256sum benign_sample.txt
f8e0bc72ac2e3e445c6ef5f1f6c2acc9b3fe16e1758f33d25a8535fec7f6d6 benign_sample.txt
kali@kali:~/ApexPlanet/Task-4/malware_lab$
```

9. Risk and Mitigation Summary

Area	Evidence-based observation	Mitigation
FTP	vsftpd 2.3.4 exposed; VSFTPD backdoor module identified	Upgrade/remove; restrict exposure; secure file transfer
SSH	Legacy algorithms enumerated	Disable deprecated algorithms; update OpenSSH
Services	Multiple network services exposed	Disable unnecessary services; restrict access
Firewall	Firewall state assessed	Apply least-privilege inbound rules
Patching	Patch state assessed	Apply security updates and re-scan
Phishing	Awareness simulation created	Train users to verify senders, links and requests
Malware	Benign sample analyzed statically	Use isolated sandboxes for real samples

10. Evidence Index

The report includes the screenshots supplied for the assessment. Filenames are retained to make the report traceable to the original evidence archive.

01-Metasploitable2-Recon-Scan.png
02-Metasploit-Console.png
03-Vulnerability-Assessment.png
03-Vulnerability-Scan-Overview.png
04-Vulnerability-Finding-1.png
05-Vulnerability-Finding-2.png
06-Vulnerability-Finding-3.png
07-Vulnerability-Finding-4.png
08-SSH-Algorithm-Output.png
08-SSH-Security-Assessment.png
09-Firewall-Current-Status.png
10-Listening-Services.png
11-Patch-Assessment.png
13-Phishing-Awareness-Simulation.png
14-Benign-Sample-Static-Analysis.png
14-Metasploit-Vsftpd-Module.png
15-Benign-Sample-Analysis-Hash.png
15-Vsftpd-Module-Info.png
16-Vsftpd-Exploit-Options.png

11. Conclusion

The controlled lab assessment demonstrates a practical workflow from reconnaissance through vulnerability identification, security configuration review, awareness simulation, and basic malware-analysis concepts. The evidence most strongly supports the identification of the vulnerable VSFTPD 2.3.4 service and the need for system hardening through patching, service reduction, firewall restrictions, and modern SSH configuration.

This report intentionally distinguishes observed evidence from actions that were not successfully demonstrated in the captured screenshots.